

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 07

Design Specifications

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	7.0 - Design Specifications (24 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Topic-level mapping to the 24-slide Lecture 07 deck. The deck is credited only for content it actually shows: layout, spacing, airflow/opening logic, enclosure, and climate-specific microclimatic solutions. Questions requiring detailed envelope-component analysis or full solar-shading geometry are marked Partial and handed off to Lectures 08-09.

RESULT

Lecture 07 is the form-and-airflow foundation of Set A3. Its strongest historical matches are the 2018 layout/spacing/opening question, the repeated 2021-2023 hot-dry vs warm-humid spacing questions, and the 2023 natural/cross-ventilation question. Slides 18-23 are the highest-return block because they directly connect openings, enclosure and climate-specific form to exam sketches.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 07 moves from site/building layout to internal spacing, airflow, enclosure and climate-specific design. It contains a large number of sketch-based design cues: broad spacing for breeze penetration, staggered blocks, opening position and size, cross ventilation, enclosure material/form/insulation, and contrasting warm-humid versus hot-dry responses. The deck also uses Parekh House and other images as visual examples rather than as standalone theory.

24	13	5	1	7	155
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2-6	Design specification + layout/form	2018 Q2(b); 2023 Q6(b)	High-value planning block: layout, breeze penetration, staggered blocks, wind angle and compactness.
7-15	Spacing + internal-space examples	2018 Q2(b); supports 2021-2023 spacing questions	Use the principles, not the project photographs, as the exam answer.
16-17	Microclimatic solutions	2021 Q7(b); 2022 Q7(a); 2023 Q7(b)	Direct hot-dry / warm-humid sketch pair.
18	Air movement: opening position, size, control, cross ventilation	2018 Q2(b); 2023 Q6(b)	One of the most exam-useful slides in the deck. Reproduce airflow arrows.
19-20	Enclosure + thermal capacity	Partial support to envelope questions	Useful bridge to Lecture 08, but not a complete component-by-component envelope answer.
21-23	Warm-humid vs hot-dry building response	2021-2023 spacing family; 2024 traditional-form application partial	High-return climate-specific design comparison with sketches.
24	Closing slide	No direct question	No study time needed.

Priority note

Study Slides 16-23 first, then 3-6. Those slides carry the strongest repeated-question value: spacing, opening placement, cross ventilation, enclosure logic and warm-humid/hot-dry design response.

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2018	Q2(b)	20	4	Discuss layout, spacing and opening position for ensuring air movement and ventilation.	Exact	3-6, 18	The deck directly shows broad spacing, staggered layout, wind-angle logic, opening position/size/control and cross ventilation.
2018	Q3(a)	22	4	Explain how climatic factors affect building-envelope components while designing microclimate.	Partial	19-20	Enclosure materials, form, insulation and thermal capacity are introduced, but the detailed roof/floor/wall/window/door analysis belongs to Lecture 08.
2018	Q3(c)	5	4	Mention the design tools for moderating microclimate.	Partial	2, 18-20	The lecture gives design-specification categories and airflow/enclosure controls, but the explicit six-item "design tools" list appears in Lecture 08.
2021	Q7(b)	10	7	Discuss microclimatic solutions for hot-dry and warm-humid climates in terms of spacing.	Exact	16-17, 21-23	The deck provides contrasting spacing/form sketches and warm-humid versus hot-dry design principles.
2021	Q8(b)	18	7	Explain how climatic factors affect the building envelope; use sketches.	Partial	19-20	General enclosure and thermal-capacity guidance is present, but the complete component-by-component response is in Lecture 08.
2022	Q7(a)	10	9	Discuss microclimatic solutions for hot-dry and warm-humid climates in terms of spacing.	Exact	16-17, 21-23	Direct repeat of the climate-specific spacing and microclimatic-solution block.
2022	Q8(b)	14	9	Explain the effects of climatic factors on the building envelope.	Partial	19-20	The deck supports enclosure material/form/insulation logic but not the full roof-floor-wall-window-door breakdown.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2023	Q6(b)	8	11	Explain natural ventilation, spacing between blocks and methods of cross ventilation.	Exact	4-5, 17-18	Broad spacing, staggered blocks and cross-ventilation/opening sketches directly answer the task.
2023	Q7(b)	10	11	Discuss microclimatic solutions for hot-dry and warm-humid climates in terms of spacing.	Exact	16-17, 21-23	Direct one-to-one match with the hot-dry/warm-humid response slides.
2023	Q8(d)	12	12	Explain how climatic factors affect the building envelope.	Partial	19-20	General enclosure guidance only; use Lecture 08 for a full answer.
2024	Q4(b)	13	14	Explain climatic factors affecting building design and how they influence it.	Partial	3-6, 18-23	The deck translates wind, form, openings, enclosure and climate response into design, but it does not cover the full climatic-data list.
2024	Q8(a)	12	15	From two traditional built-form sketches, identify the climates and analyze components, surroundings and orientation.	Direct	21-23	The warm-humid and hot-dry design principles provide a strong interpretation framework, though the exact exam sketches must still be read.
2024	Q8(b)	10	15	Explain building-envelope response in microclimate design.	Partial	19-20	The envelope is introduced at enclosure level; detailed component mechanisms belong to Lecture 08.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
S	Hot-dry vs warm-humid microclimatic solution in spacing/layout	2021, 2022, 2023	16-17, 21-23	Three consecutive papers; highly sketchable and easy to structure as a comparison.
S	Layout, spacing, openings and cross ventilation	2018, 2023	3-6, 18	Core design-specification family; supports both theory and sketch questions.
A	Building-envelope response - general enclosure support	2018, 2021, 2022, 2023, 2024	19-20	Frequent family, but Lecture 07 is only a bridge; full marks require Lecture 08.
A	Warm-humid design principles	2021-2024 related	21-22	Large openings, shading, low heat storage, reflective surfaces, double roofs, vegetation.
A	Hot-dry design principles	2021-2024 related	16, 23	Protected external space, hot-dust shielding, evaporation and reduced heat admission.

Quick recognition rules

If the question says...	Go to slides	Recognition rule
"layout / spacing / opening position / ventilation"	3-6, 18	Answer with site-block arrangement first, then inlet/outlet position and airflow path.
"hot-dry vs warm-humid spacing"	16-17, 21-23	Hot-dry = protected/compact + evaporative cooling; warm-humid = open + wind access + shade.
"cross ventilation"	18	Draw inlet and outlet on opposite/pressure-different sides with a clear occupied-zone airflow path.
"enclosure / insulation / thermal capacity"	19-20	Use material, degree of enclosure, form/insulation and climate; then hand off detailed components to Lecture 08.
"traditional forms - identify climates"	21-23	Read for ventilation openness vs protected thermal-mass/evaporative response.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	What the slides teach	Past-question links	Study action
2	Five design-specification headings	2018 Q2(b) partial	Use as the answer framework.
3-6	Layout, breeze penetration, block staggering, compactness	2018 Q2(b); 2023 Q6(b)	Practice one plan with wind arrows and one staggered-block sketch.
7-15	Internal spacing / functional distribution / examples	2018 Q2(b) support	Extract principles; do not memorize project photographs.
16-17	Microclimatic solutions - hot-dry and warm-humid	2021 Q7(b); 2022 Q7(a); 2023 Q7(b)	Reproduce both sketches and label wind, shade, pool/evaporation.
18	Air movement and opening control	2018 Q2(b); 2023 Q6(b)	Must-practice cross-ventilation diagram.
19-20	Enclosure and thermal capacity	Envelope questions - partial	Use as intro, then continue with Lecture 08.
21-22	Warm-humid building design	2021-2023 spacing; 2024 Q8(a) support	Memorize six bullets + vegetation sketch.
23	Hot-dry building design	2021-2023 spacing; 2024 Q8(a) support	Memorize protected-space / heat-storage / evaporative-cooling logic.
24	Closing	None	Skip.

5. Coverage Gaps and Diagnostic Exclusions

Lecture 07 is rich in design principles but deliberately broad. It should not be used as a substitute for the more detailed envelope and shading lectures that follow.

Gap	Affected questions	Action
Detailed roof/floor/wall/window/door climatic mechanisms and design solutions	2018 Q3(a); 2021 Q8(b); 2022 Q8(b); 2023 Q8(d); 2024 Q8(b)	Use Lecture 08 for the complete envelope answer.
Horizontal/vertical/combined shading definitions and effectiveness	2016 Q7(c), Q8(a); 2021/2022 Q8(a)	Use Lecture 09. Lecture 07 only lists "Shading Device" as a design-specification category.
Seasonal overhang geometry and solar-shading numerical	2023 Q6(a), Q7(a); 2024 Q5(c), Q7(b)	Use Lecture 09 for concept and the dedicated analytical method for numerical geometry.
Formal site-climate data collection / urban climate analysis	Site-climate questions	Use Lecture 06 / Section B site-climate material; Lecture 07 assumes design inputs are already known.

Mapping discipline	Image examples are treated as explanatory support, not as extra theory. A question is marked Exact only when the required design rule is visible in the deck; repeated envelope questions remain Partial here because the component-level mechanisms are not in Lecture 07.
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6. Recommended Study Plan

Treat Lecture 07 as a sketch-driven lecture. Revise the principles in the same order a designer would use them: layout -> spacing -> openings/airflow -> enclosure -> climate-specific response.

Step	Study block	Slides	Practice immediately
1	Hot-dry vs warm-humid microclimatic solutions	16-17, 21-23	2021 Q7(b); 2022 Q7(a); 2023 Q7(b)
2	Opening position / cross ventilation	18	2018 Q2(b); 2023 Q6(b)
3	Layout and block spacing	3-6	2018 Q2(b); 2023 Q6(b)
4	Enclosure / thermal capacity	19-20	Use as intro for 2018/2021/2022/2023/2024 envelope questions, then continue Lecture 08
5	Traditional-form climate recognition	21-23	2024 Q8(a)

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	Staggered blocks + breeze penetration	Draw wind arrows and show why alignment matters.
A	Cross-ventilation opening sketch	Show inlet, outlet, occupied zone and airflow path.
A	Warm-humid vs hot-dry comparison	Two-column table + one sketch for each climate.
B	Enclosure headings	Material, degree, form/insulation, local climate + thermal-capacity note.

Bottom line

Lecture 07 gives the design grammar of Set A3. If Slides 16-23 are mastered with sketches, the repeated spacing/ventilation family becomes a reliable scoring area; use Lectures 08-09 to complete envelope and shading questions.